

Pedestrian Evacuation Time Estimator

Workflow

Tsunami's are a great threat that are often ignored until it is too late. The tool created is one to help planners to estimate time for pedestrians to evacuate in the event of a tsunami. This tool takes into account different walking speeds based on slope and land cover. Times are then broken up into 5 minute increments to showcase the different cut off points to get to one of the assembly points.

The tool only requires a DEM, land cover raster, assembly points, critical facility points, tsunami free polygon, and a CSV. The CSV file breaks down the different walking speeds over different land cover types. The code starts by naming those parameters to be used in the tool, including optional outputs created in the analysis process. Setting up the environment to make sure the analysis is more efficient by masking the extent of the data being used is then coded in. The data is masked using the DEM selected.

Step 0 would be the rasterization to memory the tsunami free polygon, which will be used later in the analysis, but already set to be the same as the DEM raster. The actual first step then is to calculate the slope from the DEM in percent and convert the percent over to fraction. Fractions are needed because that's how the values are needed in the Tobler's Hiking Function. The Tobler's Hiking Function is used to take into account the speed people walk depending on the elevation they are going through. The function is a known concept showcasing how fast people can walk, with a maximum speed being reached when traveling on a -5% slope. An optional raster file can be saved showing how much slope impacts the speed of travel over the extent of the DEM.

Step 2 reclassifies the land cover to walking speed using a CSV. People in an emergency situation rarely follow all the rules, and most likely will go any way they can to get to safety. The CSV file created speeds for how people travel over different land cover types. The speeds were created based on the US Army Corps of Engineers, DOGAMI, and other transportation studies to determine how fast the average individual travels through different land cover types. Using this information, the land cover raster was reclassified to speed, which can be saved to compare between Tobler's Hiking Function. The next step was to combine both the land cover speed and Tobler's Hiking Function, keeping only the slowest of the two in a new raster, if desired. The

combined speed raster is then used to create the cost raster. The cost raster is created by inverting the combined speed raster, going from m/min to min/m allowing time to be the part taken into account moving forward.

The Cost Distance analysis is done next, which accumulates time to get to the assembly points. The raster created had to be restricted to be only between 0 to 120 minutes, saved as `time_evac_safe`, in order to speed up the process and understand what the chances truly are to get to one of these assembly points. This process also created a backlink, which gives directionality, used in the next step in creating a least cost polyline. The Least Cost Path uses critical facilities, namely police stations, schools, hospitals, and fire stations, as starting points to get to assembly points. The thinking here is that most people in the event of an emergency would gravitate towards those facilities for guidance and knowing the paths people should take from those locations would be beneficial. These were created as a raster, but converted and saved as polylines.

Lastly, an isochrones is created based only on travel time away from the assembly points. These are scores in descending order from 9 to 2, every 5 minutes. This is then saved to memory as the process continues. The `cost_raster` is being used as a representation of all the landmass that isn't water. These two rasters are then combined keeping the values of the isochrones, but also labeline the extent not covered by the isochrones as a 0. This 0 is supposed to indicate an extremely low chance of making it to one of the assembly points within two hours. Lastly, the raster of the tsunami free areas are given a value of 10 and placed on top of the previously combined rasters. This final raster showcases the chance of getting to one of these assembly points or being in one of the safe locations already.

Instructions

Open the package. All the files needed are in the input folder. The parameters and the corresponding files to select as follows:

Label on Tool	Input File Names
DEM	DEM_10m.tif
Land Cover	NLCD_2016_Land_Cover_OR.tif
Land Cover Speed Table	Speed.csv

Assembly Locations	Assembly_areas.shp
Critical Facilities	CriticalFacilities2.shp
Tsunami Free Zone	OutsideHazardArea_GreenHighGround.shp

The tool takes a very long time to run, but there are messages along the way to make sure the program didn't just freeze and is actually working.